

①

ASSIGNMENT - 4

Decrypt the following ciphertext using the Hill cipher

1. Cipher text : HLDTRIHJDXWQCMAG

$$\text{Key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

use (A=0, B=1, ..., Z=25)

Find the plain text

step 1: Find determinant of K

$$\det(K) = 3(7) - 5(2) = 21 - 10 = 11$$

step 2: Find K^{-1}

$$K^{-1} = (\det K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$11^{-1} \pmod{26} : 26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 \mp 3(-1) = 1$$

$$26 + 11(-2) + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 \mp 7 = 19$$

$$11^{-1} \pmod{26} = 19 \pmod{26} = 19$$

$$\therefore K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

②

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

step 3 : Divide ciphertext into pairs

HL DT QI HJ DX WQ CM AG

$$\text{Formula} = P = K^{-1}C \text{ mod } 26$$

$$HL = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{matrix} T \\ D \end{matrix}$$

$$DT = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{matrix} P \\ S \end{matrix}$$

$$QI = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{matrix} E \\ C \end{matrix}$$

$$HJ = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{matrix} R \\ E \end{matrix}$$

$$DX = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = \begin{matrix} T \\ M \end{matrix}$$

$$WQ = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 290 \\ 278 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{matrix} E \\ S \end{matrix}$$

$$CM = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 18 \\ 0 \end{bmatrix} = \begin{matrix} S \\ A \end{matrix}$$

$$AG = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{matrix} G \\ E \end{matrix}$$

$\therefore$ Plaintext = TOP SECRET MESSAGE

2. Consider the Hill cipher key matrix

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$$

Cipher text: RXAODY

Find the plaintext for the above ciphertext

step 1: Find determinant of K

$$\det(K) = 24(14) - 19(5) = 336 - 95 = 241$$

$$241 \bmod 26 = 7$$

step 2: Find K^{-1}

$$K^{-1} = (\det K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$(\det K)^{-1} = 7^{-1} = 26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$[26 + 7(-3)] + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} \bmod 26 = 15 \bmod 26 = 15$$

$$\therefore K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} \pmod{26}$$

④

$$K^{-1} = \begin{bmatrix} 210 & -285 \\ -75 & 360 \end{bmatrix} \text{mod } 26$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

step 3: Devide cipher text into pairs:

RX AO DY

Formula: $P = K^{-1}C \text{mod } 26$

$$RX = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 57 \\ 557 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{matrix} F \\ L \end{matrix}$$

$$AO = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} O \\ W \end{matrix}$$

$$DY = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{matrix} E \\ R \end{matrix}$$

FLOWER - Plaintext

3. Confusion VS Diffusion

Analyze the following classical ciphers with respect to confusion & diffusion. complete the table & justify your classification based on how each cipher transforms the plaintext

Cipher	Does it change the character/symbol value	Does it change the character position?	Confusion / Diffusion	Justification
Shift Cipher	Yes	No	Confusion	Each plaintext character is replaced by another char by shifting its value. The position remains unchanged
Multiplicative Cipher	Yes	No	Confusion	Each char value is multiplied by a key modulo 26. Char position remain unchanged
Affine Cipher	Yes	No	Confusion	Each char is transformed using $C = (aP + b) \bmod 26$. The symbol value changes but its position does not
Transposition Cipher	No	Yes	Diffusion	The char themselves remain unchanged, but their positions are rearranged.

4.

Based on your observations, explain the relationship between substitution & confusion & between transposition & diffusion.

• Relationship between substitution & Confusion.

* Substitution ciphers such as shift, multiplicative & Affine ciphers replace one symbol with another.

* Therefore, they primarily provide confusion by making the relationship between the plaintext symbols & ciphertext symbols less direct.

• Relationship between transposition & Diffusion.

* A transposition cipher does not change the actual char instead, it rearranges their positions.

⑥

* This spreads the plaintext char across different positions in the ciphertext & therefore provides diffusion.

- substitution $\rightarrow$ confusion
- Transposition $\rightarrow$ Diffusion

5. Transposition

Encryption

encrypt the message "The arrow is pointing north direction" using

- simple with key = 4

$\rightarrow$ Key = 4

$\therefore$ No. of columns = 4

T	H	E	A
R	R	O	W
I	S	P	O
I	N	T	I
N	G	N	O
R	T	H	D
I	R	E	C
T	I	O	N

Ciphertext = TRIINRITHRSN4TRIEOPTNHEDAWOIODGN

- Keyword with Keyword = ATLAS

Keyword length = 5

$\therefore$ No. of columns is 5

⑦

A ₁	T ₅	L ₃	A ₂	S ₄
T	H	E	A	R
R	O	W	I	S
P	O	I	N	T
I	N	G	N	O
R	T	H	D	I
R	E	C	T	I
O	N	X	X	X

Ciphertext = TRRIRROHOONTENEWIGHCXAINNDTXRSTOIX

double

Key 1 = BREAK

Key 2 = hog

Key 1 = BREAK, length = 5, No. of columns = 5

B ₂	R ₅	E ₃	A ₁	K ₄
T	H	E	A	R
R	O	W	I	S
P	O	I	N	T
I	N	G	N	O
R	T	H	D	I
R	E	C	T	I
O	N	X	X	X

Initial ciphertext = AINNDTXTRRIRROEWIGHCXRSTOIXHOON
-NTEN

Key 2 = LOG, length = 3, No. of columns = 3

2 3 1
 L O G
 A I N
 N D T
 X T R
 P I R
 R O E
 W I G
 H C X
 R S T
 O I T
 X H O
 O N T
 E N X

∴ Final ciphertext is

NTRRE4XTIOTXANXPRWHKROXOEIDT
 IOICSIHNN

Decryption

- simple with key 10

TIDIE MRNME REOQO SANOW RSLTP EEEDOSSNU
 AHTUD BIENP GODAE PEIXD ELNSX AHTUD

→ string length = 60, Key 10

∴ $60/10 = 6 \Rightarrow$ No. of rows = 6

T R O O P S H E A D
 I N G W E S T N E E
 D M O R E S U P P L
 I E S S E N D G E N
 E R A L D U B O I S
 M E N T O A I D X X

(9)

∴ Plaintext = TROOPS HEADING WEST NEED MORE
SUPPLIES SEND GENERAL DUBOIS
MEN TO AID

Keyword

SECRET

Ciphertext: TLEDI NHPTR SOIIT MAXSA XERXT ANONI
SNFOT X

Key = ~~SECRET~~ SECRET. Key length = 6. No. of columns = 6

Ciphertext length = 36 ∴ No. of rows = $36/6 = 6$

SECRET

5 2 1 4 3 6 → Numbering of Keyword.

S E C R E T

T H I S

A P L A I N

T T E X T F

O R D E M O

N S T R A T

I O N X X X

∴ Plaintext : THIS IS A PLAINTEXT FOR
DEMONSTRATION

- Keyword

Keyword - PLAIN

⑩

CIPHER - VEY EAX ARA ATX HQD

Keyword = PLAIN, length = 5, No. of columns = 5

Ciphertext length = 15, No. of rows = $15/5 = 3$

P L A I N

H A V E A

G R E A T

D A Y

∴ Plaintext is

HAVE A GREAT DAY

Permutation

Find the inverse permutation for

1. 4 3 1 5 2

Position

1 2 3 4 5

P-Box

4 3 1 5 2

∴ Inverse Permutation

3 5 2 1 4

∴ Inverse Permutation, $P^{-1} = (3, 5, 2, 1, 4)$

⑪

2. 7 4 1 3 5 2 6

Position	1	2	3	4	5	6	7
P-box	7	4	1	3	5	2	6

$\therefore$ Inverse Permutation, $P^{-1} = (3, 5, 2, 1, 4, 7, 6)$

3. 4 1 3 2

Position	1	2	3	4
P-Box	4	1	3	2

Inverse permutation, $P^{-1} = (2, 4, 3, 1)$

4. 3 5 1 4 2

Position	1	2	3	4	5
P-box	3	5	1	4	2

Inverse Permutation, $P^{-1} = (3, 5, 1, 4, 2)$